

Main Report — HW06 API Testing

MSSV: 23127044

Assignment: HW06 — API Testing (Postman + Newman + AI pipeline)

Date: 2026-08-29

SUT: HW2/testing-seminar-eshop-sut (localhost:3000)

1. Executive Summary

Bài HW06 triển khai pipeline kiểm thử API đầy đủ cho 3 feature giữ nguyên từ HW02:

API	Feature	Endpoint	TC executed
1	FR-04 Profile	PUT /api/users/me	44
2	FR-10 Order State	PUT /api/admin/orders/:id/status + user cancel	44
3	FR-18 Admin Orders	GET /api/admin/orders	44

Mỗi API đi qua 5 bước: **Generate (AI step-by-step)** → **Audit (VALID/INVALID/INCOMPLETE)** → **Extend (≥6 TC)** → **Execute (Postman/Newman)** → **Report (bugs)**.

Header bắt buộc X-Student-Id: 23127044 được gắn ở collection pre-request script.

2. API 1 — FR-04 Profile

2.1 Generate (6 AI sessions)

Session	Prompt focus	Output
1	5 input variables + partitions (name, phone, address, role, JWT)	20 domain TC
2	BVA phone regex <code>^[1-9][0-9]{8,9}\$</code> ON/OFF/IN/OUT	8 boundary TC
3	SEC-06 role=admin in body	3 security TC
4	SEC-02 auth: missing/expired/malformed token	5 auth TC
5	Schema: response keys after PUT	3 schema TC
6	Merge	38 TC → api1-profile-generated.csv

2.2 Audit

Đối chiếu fr04-profile/domain-testing.md + bva.md :

- 6 TC INVALID — AI giả định server validate phone/name như frontend
- Sửa expected status theo server.js thực tế

2.3 Extend (+6 TC)

TC	Lý do AI miss
API1-E-01	SEC-06 abstract, không đọc server cho phép update role
API1-E-02	Regex frontend ≠ business rule VN phone
API1-E-03	HTML required ≠ backend validation

API1-E-04	Auth negative path thiếu
API1-E-05	SQLi stability assert
API1-E-06	Invalid role enum

2.4 Execute

- Collection: 23127044_API1_Profile.postman_collection.json
- Newman: results/api1/newman-report.html

2.5 Bugs

BUG-A1, BUG-A2, BUG-A3 — xem bug-report.md

3. API 2 — FR-10 Order State

3.1 Generate

State machine từ fr10-order-state/domain-testing.md :

- Valid transitions: pending→confirmed→shipping→delivered, cancel paths
- Invalid: skip-step, backward, terminal states
- User cancel sub-endpoint
- 38 TC generated

3.2 Audit

- BUG-B1/B3 TC giữ expected đúng nghiệp vụ (400)
- SEC-03 user→admin API: sửa expected 403→200 (server thiếu role check)

3.3 Extend (+6 TC)

Focus invalid transitions và user vs admin cancel rules.

3.4 Execute

Setup chain: admin login → user login → cart → checkout → status prep per from_status .

4. API 3 — FR-18 Admin Orders

4.1 Generate

Partitions admin_token từ fr18-order-admin/domain-testing.md :

- Admin/user/no/expired token
- Schema array + required fields
- Empty vs populated list
- 38 TC generated

4.2 Audit

4 TC SEC-03 corrected — backend trả 200 cho user token.

4.3 Extend (+6 TC)

Empty list, all fields schema, Content-Type header.

5. AI Misinterpretation Examples (≥3)

#	AI said	Actual Newman	Correction
1	Phone 0912345678 → expect 400 (invalid regex)	Server returns 200 (no validation)	Expected = 200 for API layer; BUG-A1 is frontend-only
2	role=admin in body → expect 200 (profile update success)	Should be 403 SEC-06	Expected = 403; documents BUG-A2
3	User token on GET /api/admin/orders → expect 403	Server returns 200 with full order list	Expected = 200 per actual; TC documents missing role check
4	canceled → delivered → expect 400	Server returns 200 (BUG-B1)	Keep expected 400; Newman FAIL = bug evidence

6. Postman Features Used

Xem `docs/postman-features-used.md` :

- Workspace, Collections ×3, Environment variables
- Pre-request scripts (X-Student-Id, conditional login)
- Test scripts (status, schema, JSON path)
- Collection Runner + CSV data-driven
- Newman CLI + htmlextra reports

7. CI/CD

GitHub Actions workflow `.github/workflows/api-tests.yml` — xem `docs/cicd-report.md` .

8. Agent Skill

`agent-skills/api-test-generator-skill/SKILL.md` — pipeline tái sử dụng cho API spec → Postman TC.

Diagram: `docs/test-generator-diagram.mmd` → user vẽ PNG tay.

9. References

- HW02 FR-04/10/18 domain testing artifacts
- API spec: `HW2/testing-seminar-eshop-sut/api_specification.md`
- SEC requirements: SEC-01 → SEC-07